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Absolute Value

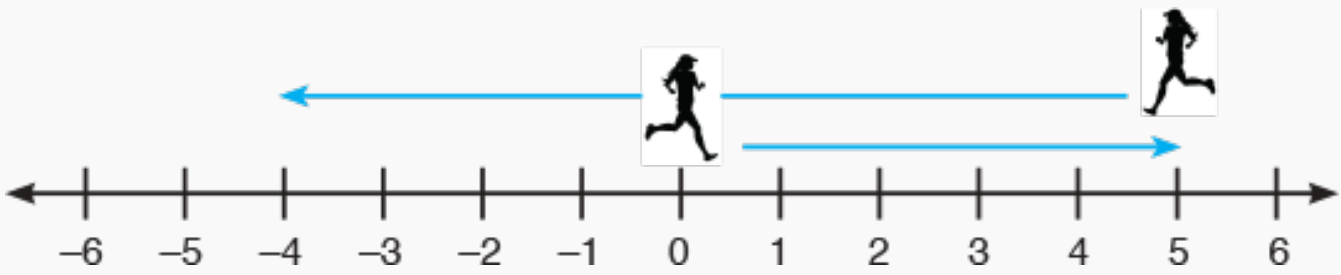
Toolbox

Absolute Value

Example 1

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A jogger runs 5 kilometers in a positive direction, turns around, and runs 9 kilometers in the opposite direction. How many kilometers did the jogger run in all?



The problem seems to say that you should add $+5$ and -9 , which equals -4 . But that answer doesn't make sense. How can anyone run a negative number of kilometers?
If you remove the number line, the problem is easy. The jogger ran 5 kilometers and then 9 kilometers: $9 + 5 =$ **14 kilometers**

This problem shows the need for absolute value in problem solving. The **absolute value** of a number is its distance from zero. To show the absolute value of a number, write the number inside vertical bars.

The absolute value of 3 is 3. $|3| = 3$

The absolute value of -3 is 3. $|-3| = 3$

$|3| = |-3|$ because both are the same distance from 0.

You can use absolute value to describe the actions of the jogger mentioned at the start of the lesson. First, the jogger ran $|5|$ kilometers. Then he turned around and ran $|-9|$ kilometers.

$|5| + |-9| = 5 + 9 = 14$ kilometers

Study how absolute value is applied in these examples.

Example 2

Hide

Find the value of $|3 - 10|$.

Step 1

Perform the operation inside the absolute value bars. $|3 - 10| = |-7|$

Step 2

Find the absolute value of the number in the bars. $|-7| = 7$
The value of $|3 - 10|$ is **7**.

Example 3

Hide

Find the value of $-|-15|$.

Step 1

Perform the operation inside the absolute value bars. $-|-15| = -(15)$

Step 2

Apply the negative sign that was outside the absolute value bars. $-(15) = -15$
The value of $-|-15|$ is **-15**.

Key Terms

- absolute value
- distributive property
- equivalent expressions
- negation

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Lesson 2: Expressions

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